Stochastic Processes

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- Random Variable (随机变量): X;
- Random Vector (随机向量): $(X_1, X_2, \cdots, X_n)$.
  [(Joint) distribution.]

$$\Downarrow$$

- **Stochastic Processes (随机过程)**:

  Discrete time: $\{X_n | n \in \mathbb{N}\} = \{X_1, X_2, \cdots\}$;

  Continuous time: $\{X_t : t \in \mathbb{R}\}$.

  Think of n or t as a time parameter.

Some important stochastic processes:

1. Markov Chains (马氏链)
2. Poisson Processes (泊松过程)
3. Renewal Processes (更新过程)
4. Continuous Time Markov Chains (连续时间马氏链)
5. Martingales (鞅)
6. Brownian Motion (布朗运动)

Chapter 1 Markov Chains

1.1 Definition and Examples

Discrete Time Markov Chain: X_n is a *discrete time Markov chain* with *transition probabilities* $p(i,j)$ if

$$P(X_{n+1}=j|X_n=i, X_{n-1}=i_{n-1},\ldots,X_0=i_0)=p(i,j)$$

for any $j, i, i_{n-1},\ldots,i_0$.

Remark

- State space $\{i,j,k,\ldots\}$ is finite (or countable): transition probabilities form a **transition matrix**.
- Markov: given current, future is independent of past.
- Temporally homogeneous: parameters do not depend on time.

Markov Chain

$\Downarrow$ $\quad\quad$ $\Uparrow *$ gives rise to

Transition (stochastic) matrix $P = [p(i,j)]$:

- $p(i,j) \geq 0$ (nonnegative entries);
- $\sum_j p(i,j) = 1$ (row summing to 1).

* Intuition: think of the chain step by step.

Rigorous argument: measure theory.

Example 1.1: A gambling game

A gambling game:

- Each turn: win \$1 with probability $p = 0.4$,
  lose \$1 with probability $1 - p = 0.6$.
- Quit the game: fortune reaches 0 or \$N.

X_n: money after n plays, "Markov chain".

Transition probability:

$$p(i, i-1) = 0.6, p(i, i+1) = 0.4, \quad \text{if } 0 < i < N;$$
$$p(0, 0) = 1, p(N, N) = 1 \,.$$

E.g., $N = 5$:

$$\text{Transition matrix:}\quad \begin{array}{c} \\ \mathbf{0} \\ \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{4} \\ \mathbf{5} \end{array} \begin{array}{c} \begin{array}{cccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Examples 1.10: Two-stage Markov Chain

A basketball player makes a shot with probabilities:

$1/2$ if he has missed the last two times;

$2/3$ if he has hit one of his last two shots;

$3/4$ if he has hit both of his last two shots.

X_n: miss or hit at the n^{th} shot.

NOT Markov! (X_{n+1} depends on X_n and X_{n-1}.)

$$Y_n = (X_n, X_{n+1}): \text{ Markov, } \quad \begin{array}{c} \\ \textbf{HH} \\ \textbf{HM} \\ \textbf{MH} \\ \textbf{MM} \end{array} \begin{array}{c} \begin{array}{cccc} \textbf{HH} & \textbf{HM} & \textbf{MH} & \textbf{MM} \end{array} \\ \begin{bmatrix} 3/4 & 1/4 & 0 & 0 \\ 0 & 0 & 2/3 & 1/3 \\ 2/3 & 1/3 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \end{array}.$$

1.2 Multistep Transition Probabilities

One-step to Multistep Transition

$p(i,j) = P(X_{n+1} = j | X_n = i)$: 1-step trans. prob. $i \to j$.

Theorem

The m^{th} step transition probability

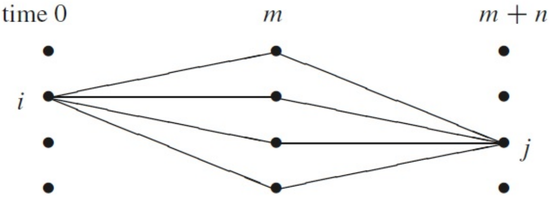
$$P(X_{n+m} = j | X_n = i) = p^m(i,j)$$

where p^m is the m^{th} power of transition matrix p.

$$p^{m+n}(i,j) = \sum_k p^m(i,k)p^n(k,j):$$

Heuristic:

- $i \to j$ in $m+n$ steps $\iff$
  $i \to k$ in m steps, then $k \to j$ in n steps;
- Markov property: the two parts are independent.



Example 1.11

Choose $N = 4$ in Example 1.1.

$$p = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \; p^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0.24 & 0 & 0.16 & 0 \\ 0.36 & 0 & 0.48 & 0 & 0.16 \\ 0 & 0.36 & 0 & 0.24 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

$$p^3, \ldots, p^{20} \ldots, \; \lim_{n\to\infty} p^n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 57/65 & 0 & 0 & 0 & 8/65 \\ 45/65 & 0 & 0 & 0 & 20/65 \\ 27/65 & 0 & 0 & 0 & 38/65 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

1.3 Classification of States

Notations & Important Quantities

Chain with a fixed initial state:

$$P_x(A) = P(A|X_0 = x) \ , \quad E_x(Y) = E(Y|X_0 = x) \ .$$

First return time to y: $T_y = \min\{n \geq 1 : X_n = y\}$.
(Hitting time, if $n \geq 0$.)

Number of visits to y (after time 1): $N(y) = \sum_{n=1}^{\infty} 1_{\{X_n = y\}}$.

Return (to y at least once) probability:

$$\rho_{yy} = P_y(T_y < \infty) = P_y(N(y) \geq 1) \ .$$

$$\rho_{xy} = P_x(T_y < \infty) = P_x(N(y) \geq 1) \ .$$

For a state y,

- y is **recurrent** if one of the following is true.

$$\rho_{yy} = 1 \Leftrightarrow P_y(N(y) = \infty) = 1 \text{ (why the name)}$$
$$\Leftrightarrow E_y N(y) = \sum_{k=1}^{\infty} \rho_{yy}^k = \infty.$$

- y is **transient** if one of the following is true.

$$\rho_{yy} < 1 \Leftrightarrow P_y(N(y) < \infty) = 1 \text{ (why the name)}$$
$$\Leftrightarrow E_y N(y) = \sum_{k=1}^{\infty} \rho_{yy}^k < \infty.$$

Example Gambler's Ruin when $N = 4$:

$$\begin{array}{c} \\ \mathbf{0} \\ \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{4} \end{array} \begin{array}{c} \begin{array}{ccccc} \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0.6 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

Recurrent: **0** (bankrupt), **4** (happy winner).

Transient: other states **1**, **2**, **3**.

Between-State Relations:

y **is accessible from** x $(x \to y)$:

$$\rho_{xy} = P_x(T_y < \infty) > 0 \quad (\Leftrightarrow \exists n, \text{ s.t. } p^n(x,y) > 0).$$

(Transitivity: $x \to y$ and $y \to z \Rightarrow x \to z$.)

x **communicates with** y $(x \leftrightarrow y)$: $x \to y, y \to x$.

Theorem

If $x \to y$ and $\rho_{yx} < 1$, then x is transient.

A set A is **closed**: $p(i,j) = 0 \ \forall i \in A, j \notin A$.
(impossible to get out from A.)

A set B is **irreducible**: $i \leftrightarrow j \ \forall i,j \in B$.

Theorem

If a set C is finite, closed and irreducible, then all states in C are recurrent.

Theorem

If the state space S is finite, then $S = T \cup R_1 \cup \cdots \cup R_k$ (disjoint union), where T is the set of transient states and R_i, $1 \leq i \leq k$, are closed irreducible sets of recurrent states.

Proof

- Define $T = \{x : x \to y, y \not\to x \text{ for some } y\}$.
- Pick $x \in S - T$, let $C_x = \{y : x \leftrightarrow y\}$ and $R_1 = C_x$.
- If $S = T \cup R_1$, Stop. Otherwise, pick $w \in S - T - R_1$, repeat. Will stop after finite iterations. $\square$

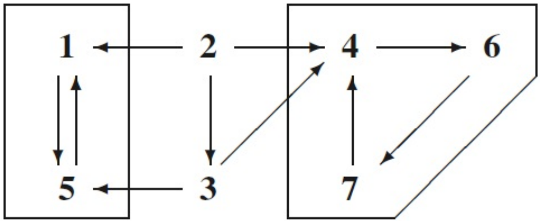
Remark No finiteness, no recurrent guaranteed.

Example 1.14 A 7-state Chain

$$
\begin{array}{c c}
 & \begin{array}{ccccccc} \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{4} & \mathbf{5} & \mathbf{6} & \mathbf{7} \end{array} \\
\begin{array}{c} \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \\ \mathbf{4} \\ \mathbf{5} \\ \mathbf{6} \\ \mathbf{7} \end{array} &
\begin{bmatrix}
0.7 & 0 & 0 & 0 & 0.3 & 0 & 0 \\
0.1 & 0.2 & 0.3 & 0.4 & 0 & 0 & 0 \\
0 & 0 & 0.5 & 0.3 & 0.2 & 0 & 0 \\
0 & 0 & 0 & 0.5 & 0 & 0.5 & 0 \\
0.6 & 0 & 0 & 0 & 0.4 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.2 & 0.8 \\
0 & 0 & 0 & 1 & 0 & 0 & 0
\end{bmatrix}
\end{array}
$$

Draw a graph containing $i \to j$ if $p(i,j) > 0$ and $i \neq j$.

Graph without self-loops:



Conclusion:

- $T = \{\mathbf{2}, \mathbf{3}\}$: transient states.
- $R_1 = \{\mathbf{1}, \mathbf{5}\}$, $R_2 = \{\mathbf{4}, \mathbf{6}, \mathbf{7}\}$: recurrent states.

1.4 Stationary Distributions

1.6 Special Examples

Stationary Distribution

Initial distribution: $\mathbf{q}=(q(i))_{\{i\in S\}}$, i.e., $P(X_0=i)=q(i)$.
(represented by a row vector.)

Then, distribution at time n: $(P(X_n=i))_{\{i\in S\}}=\mathbf{q}P^n$.

Stationary Distribution: a distribution π s.t. $\pi p=\pi$.

Remark

- Distribution of X_0: $\pi \Rightarrow$ distribution of X_n: π. (Reason for the name 'stationary'.)
- π is the (normalized) right eigenvector of p corresponding to eigenvalue 1.

Existence and Uniqueness

Theorem

Suppose that the $k \times k$ transition matrix p is irreducible. Then there is a unique solution to $\pi p = \pi$ with $\sum_x \pi_x = 1$ and we have $\pi_x > 0$ for all x.

Remark

- Finite, irreducible chain $\Rightarrow$ $\exists!$ stationary distribution with > 0 entries.
- No irreducibility: existence ($\checkmark$); uniqueness (?).

Special Cases: Stationary Distr. Solving $\pi P = \pi$

- Doubly stochastic: $\sum_x p(x, y) = 1$.

  Stationary distribution: $\pi_x = \frac{1}{N}$ (uniform).

  Examples see $ 1.6 in the book.

- Detailed Balance Conditions: $\pi_x p(x, y) = \pi_y p(y, x)$.
  - DBC $\Rightarrow$ Stationarity.

    ($\pi p = \pi$: $\sum_{y \neq x} \pi_y p(y, x) = \sum_{x \neq y} \pi_x p(x, y)$.

    probability out =in, balanced condition.)
  - Such chains, starting from stationary distribution, are also called reversible Markov chains.

Example 1.29 (Not DBC)
$$\begin{array}{cc} & \begin{array}{ccc} \mathbf{1} & \mathbf{2} & \mathbf{3} \end{array} \\ \begin{array}{c} \mathbf{1} \\ \mathbf{2} \\ \mathbf{3} \end{array} & \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.3 & 0.1 & 0.6 \\ 0.2 & 0.4 & 0.4 \end{bmatrix} \end{array}.$$

Doubly Stochastic $\Rightarrow$ uniform stationary distribution.
But, no detailed balanced condition.

Example 1.30 (Birth and Death Chains are DBC)

State space (population): $\{l, l+1, \ldots, r-1, r\}$.

Transition probability: $p_x + q_x \leq 1$,

$$p(x,x+1)=p_x, x<r \ ; \quad p(x,x-1)=q_x, x>l \ ;$$
$$p(x,x)=1-p_x-q_x, l\leq x\leq r \ ; \quad p(x,y)=0, \text{others}.$$

DBC $\Rightarrow \pi_{l+i} = \pi_l \frac{p_{l+i-1}\cdots p_{l+1}p_l}{q_{l+i}\cdots q_{l+2}q_{l+1}}$, $\forall 1 \leq i \leq r-l$.

(Stationary measure, normalized to stationary distr..)

(For concrete birth-death chain examples, see §1.6)

Example 1.33 (Random Walks on Graphs: DBC)

A graph $G = (V, E)$: V vertex set; E edge set.
Adjacency matrix A: $A(u, v) = 1$ or 0, $uv \in$ or $\notin E$.
Degree of vertex u (# neighbors): $d(u) = \sum_v A(u, v)$.

Random walks on graphs:

Markov chain with state space V,
Transition: $p(u, v) = A(u, v)/d(u)$,
(move equally likely to one neighbor).

$d(u)$ satisfies DBC $\Rightarrow$ normalized to $\pi_u = \frac{d(u)}{\sum_u d(u)}$.

1.5 Limit Behavior

Convergence (Ergodic) Theorem: Limiting Distributions

$$\lim_{n\to\infty} p^n(x,y) \ ?$$

- Limit: must be stationary, if exists.

$$\lim_{n\to\infty} p^n(x,\cdot) = \left(\lim_{n\to\infty} p^{n-1}(x,\cdot)\right) p.$$

Example (no limit) $\quad p(\mathbf{0},\mathbf{1}) = 1,\ p(\mathbf{1},\mathbf{0}) = 1.$

$$p^{2k}(\mathbf{0},\mathbf{0})=1,\ p^{2k+1}(\mathbf{0},\mathbf{0})=0 \text{ (even-odd parity).}$$

Periodicity

Period of a state x: greatest common divisor (g.c.d.) of $I_x = \{n \geq 1 : p^n(x,x) > 0\}$.

- $i, j \in I_x \Rightarrow i + j \in I_x$.
- x has period 1 $\Rightarrow \exists n_0$ s.t. $n \in I_x \; \forall n \geq n_0$.
- $x \leftrightarrow y \Rightarrow x, y$ same period.

Aperiodic chain: all states has period 1.

(common; or, chain $\rightarrow$ lazy chain (aperiodic).)

Theorem (Convergence)

Markov chain: irreducible, aperiodic, $\exists$ *a stat. distr.* π.
Then $\lim_{n\to\infty} p^n(x,y) = \pi_y$.

Asymptotic Frequency; Mean Return Time

Theorem (Asymptotic Frequency)

Markov chain: irreducible, recurrent.

$N_n(y)$*: number of visits to* y *up to time* n*. Then*

$$\frac{N_n(y)}{n} \xrightarrow{n\to\infty} \frac{1}{E_y T_y} \quad a.s. \quad .$$

Theorem (Mean Return Time)

Markov chain: irreducible, $\exists$ a stationary distribution π. Then,

$$\pi_y = \frac{1}{E_y T_y} \, .$$

Hence, the stationary distribution is unique.

Note that no aperiodicity is assumed for uniqueness.

Long Run Cost: Time Average $\to$ Space Average

Theorem (Long Run Cost of a Markov chain)

Markov chain: irreducible, $\exists$ *a stationary distribution* π.
$\sum_x |f(x)|\pi_x < \infty$. *Then*

$$\frac{1}{n}\sum_{m=1}^{n} f(X_m) \xrightarrow{n\to\infty} \sum_x f(x)\pi_x \ \ a.s. \ .$$

1.8 Exit Distributions

1.9 Exit Times

Example 1.43 (Gambler's Ruin) Consider a gambling game: win \$1 with probability p or lose \$1 with probability $q = 1 - p$. Quit if fortune reaches N or 0. Question: ruin probability? Duration of the game?

X_n: fortune at time n, Markov chain.

$$p(i, i-1) = q = 1-p, \ p(i, i+1) = p, \ 0 < i < N$$
$$p(0,0) = 1, \ p(N,N) = 1$$

$V_N = \min\{n \geq 0 : X_n = N\}, \ V_0 = \min\{n \geq 0 : X_n = 0\}$, $\tau = V_0 \wedge V_N$.

Ruin probability: $P_x(V_0 < V_N)$ (exit distribution).
Duration of the game: $E_x\tau$ (exit time).

- Ruin Probability: $h(x) = P_x(V_0 < V_N)$.

  By one-step reasoning and Markov property:

  $$h(x) = ph(x+1) + qh(x-1),\ 0 < x < N.$$

  $$h(0) = 1,\ h(N) = 0.$$

  $p = \frac{1}{2}$, $h(x) = \frac{N-x}{N}$.

  $p \neq \frac{1}{2}$, $h(x) = \frac{\theta^x - \theta^N}{1-\theta^N}$, where $\theta = q/p$.

  Remark When $N = \infty$,

  - When $p \leq \frac{1}{2}$: $P_x(V_0 < \infty) = 1$
  - When $p > \frac{1}{2}$, $P_x(V_0 < \infty) = \left(\frac{q}{p}\right)^x$.

- Duration of the game: $g(x) = E_x\tau$.

  By one-step reasoning and Markov property:

  $$g(x) = 1 + pg(x+1) + qg(x-1),\ 0 < x < N.$$

  $$g(0) = 0,\ g(N) = 0.$$

  $p = \frac{1}{2}$, $g(x) = x(N-x)$.

  $p \neq \frac{1}{2}$, $g(x) = \frac{x}{q-p} - \frac{N}{q-p} \cdot \frac{1-(q/p)^x}{1-(q/p)^N}$.

  Remark When $N = \infty$,

  - When $p < \frac{1}{2}$: $E_x V_0 = \frac{x}{q-p}$.
  - When $p \geq \frac{1}{2}$, $E_x V_0 = \infty$.
    (This includes the case $p = \frac{1}{2}$.)

Theorem (Exit Distribution)

Consider a Markov chain with state space S.

A, B: *disjoint subsets of* S; $C = S - A \cup B$: *finite.*

$P_x(V_A \wedge V_B < \infty) > 0$, $\forall x \in C$.

Suppose that $h(a) = 1$, $a \in A$; $h(b) = 0$, $b \in B$;

$$h(x) = \sum_{y \in S} p(x, y) h(y), \ x \in C.$$

Then $h(x) = P_x(V_A < V_B)$.

Theorem (Exit Time)

Consider a Markov Chain with state space S.

$A \subset S$, $C = S - A$*: finite.*

$P_x(V_A < \infty) > 0, \forall x \in C$.

Suppose that $g(a) = 0$, $a \in A$, *and*

$$g(x) = 1 + \sum_{y \in S} p(x, y) g(y),\ x \in C.$$

Then $g(x) = E_x V_A$.

Example 1.47 (Waiting for TT) average number of flipping a coin before we see two consecutive Tails.

Markov chain: $\#$Tails in a row (stop at TT).

$$\begin{array}{c c} & \begin{array}{ccc} \mathbf{0} & \mathbf{1} & \mathbf{2} \end{array} \\ \begin{array}{c} \mathbf{0} \\ \mathbf{1} \\ \mathbf{2} \end{array} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Solution $E_{\mathbf{0}} V_{\mathbf{2}} = 6.$

1.10 Infinite State Spaces

Major complication:

- Irreducible, closed $\not\Rightarrow$ recurrent.
- Stationary Distribution: not guaranteed even when irreducible, closed, recurrent.

Example 1.51 (Reflecting Random Walks) Imagine a particle that moves on $\{0, 1, 2, \dots\}$.

$$p(i, i+1) = p\ , \text{ when } i \geq 0\ ;$$
$$p(0,0) = p(i, i-1) = 1 - p\ , \text{ when } i \geq 1\ ;$$

Birth-death chain: DBC $\Rightarrow \pi(i) = \pi(0)\left(\frac{p}{1-p}\right)^i$.

Case 1 ($\mathbf{p} < \frac{1}{2}$): Stationary distribution

$$\pi(i) = \frac{1-2p}{1-p}\left(\frac{p}{1-p}\right)^{i} \text{ (geometric).}$$

Irreducible, aperiodic $\Rightarrow P(X_n = j) \to \pi(j)$.

$E_0 T_0 = \frac{1}{\pi(0)} = \frac{1-p}{1-2p} < \infty \Rightarrow$ all states are recurrent.

Case 2 ($\mathbf{p} > \frac{1}{2}$): No stationary distribution!

Previously, $P_1(T_0 < \infty) = \frac{1-p}{p} < 1 \Rightarrow$ All states: transient.
(Intuition: drift to ∞.)

$P(X_n = j) \xrightarrow{n\to\infty} 0, \forall j.$ (Again, no stationary distribution.)

Case 3 ($\mathbf{p} = \frac{1}{2}$): No stationary distribution.

Previously, $P_1(T_0 < \infty) = 1 \Rightarrow P_0(T_0 < \infty) = 1$.
Thus, 0 and other states: recurrent.

Previously, $E_1 T_0 = \infty \Rightarrow E_0 T_0 = 1 + \frac{1}{2} E_1 T_0 = \infty$.
(Although $P_0(T_0 < \infty) = 1$.)

Definition (Positive/Null Recurrence)

x is said to be **positive recurrent** if $E_x T_x < \infty$; x is said to be **null recurrent** if $P_x(T_x < \infty) = 1$ but $E_x T_x = \infty$.

x: (Null) recurrence $\Rightarrow \pi(x) = 1/(E_x T_x) = 0$.

In the previous example,

$$0 \text{ is } \begin{cases} \text{transient,} & \text{when } p > \frac{1}{2}; \\ \text{null recurrent,} & \text{when } p = \frac{1}{2}; \\ \text{positive recurrent,} & \text{when } p < \frac{1}{2}. \end{cases}$$

Theorem (Pos. Rec. & Stat. Dist.)

For an irreducible chain, the following are equivalent:

(i) *Some state is positive recurrent.*

(ii) *There is a stationary distribution π.*

(iii) *All states are positive recurrent.*

Proof (**iii**) $\Rightarrow$ (**i**): obvious!

(**i**) $\Rightarrow$ (**ii**): $E_x T_x < \infty$, so $\mu(y) = \sum_{n=0}^{\infty} P_x(X_n = y, n < T_x)$

$\xrightarrow{\text{normalize}}$ a stationary distribution.

(**ii**) $\Rightarrow$ (**iii**): irreducibility, $\pi(x) = \sum_y \pi(y) p^n(y,x)$, $\forall x, n$

$$\Rightarrow \forall x,\ \pi(x) > 0,\ E_x T_x = 1/\pi(x) < \infty.$$

Example 1.52 (Branching Processes) Consider a population. Each individual in generation n gives birth to an $i.i.d.$ number (distributed as p_k) of children and dies in generation $n+1$. Mean number of children: $\mu = \sum_{k=0}^{\infty} kp_k$.

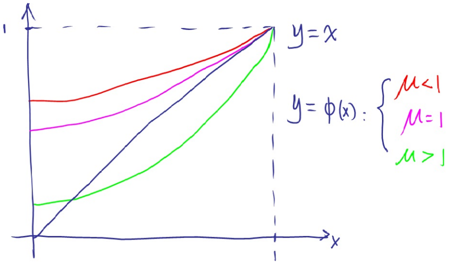
X_n: # individuals at generation n, a Markov chain. (WLOG, $X_0 = 1$)

Question: Extinction probability $\rho = P_1(T_0 < \infty)$?
Expected extinction time $E_1 T_0$?
($p_0 > 0$: otherwise, $T_0 = \infty$.)

- For extinction probability:

$$\rho = \sum_{k=0}^{\infty} p_k \rho^k := \phi(\rho).$$

Properties of $\phi(x)$: $\phi(0) = p_0$, $\phi(1) = 1$,
$\phi'(1) = \mu \geq 1$,
$\phi''(x) \geq 0$ for $0 \leq x \leq 1$.



$\boldsymbol{\mu \leq 1}$: $\rho = 1$, extinction with probability 1.

$\boldsymbol{\mu > 1}$: $\rho < 1$, all-time survival is possible.

ρ: the smallest solution of $\phi(x) = x$, $0 \leq x \leq 1$.

Proof $\rho_n = P(X_n = 0)$: P(extinction by time n).

$\cdot$ $\rho_n = \phi(\rho_{n-1})$, $\rho_n \uparrow \rho$;

$\cdot$ $\forall x_0 \in [0, 1]$, $\phi(x_0) = x_0$:

$\phi \uparrow$; $\rho_0 = 0 \leq x_0 \Rightarrow \rho_n \leq x_0$.

- For expected extinction time:

$$E_1 T_0 = \sum_{n=1}^{\infty} P_1(T_0 \geq n) = \sum_{n=1}^{\infty} P_1(X_{n-1} > 0).$$

$\boldsymbol{\mu > 1}$: $P_1(T_0 = \infty) > 0$, $E_1 T_0 = \infty$ Trivial.

$\boldsymbol{\mu < 1}$: $P_1(X_n > 0) \leq E_1 X_n = \mu^n$, $E_1 T_0 < \infty$.

$\boldsymbol{\mu = 1}$: $P(X_n > 0) \sim \frac{2}{n\sigma^2}$ (Athreya and Ney 1972),
$E_1 T_0 = \infty$.